

**Fall 2025.**

**Physics 501. Methods of Theoretical Physics I.  
Yurii Shylnov.**

**Homework # 6.**

**Due to December 2<sup>nd</sup>, 2025.**

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

**Partial Differential Equations in the cylindrical and spherical systems of coordinates.**

1. **Mark Pinsky. Section 3.3. Fourier-Bessel series for vibrating drumhead.** Problems: 3.3.6 and 3.3.7. Hint: use examples 3.3.2 and 3.3.3.
2. **Mark Pinsky. Section 3.4. An Infinite cylinder.** Problems: 3.4.1, and 3.4.2.
3. **Mark Pinsky. Sections 3.5. A finite cylinder.** Problems: 3.5.1, 3.5.2, 3.5.4, and 3.5.5.
4. **Mark Pinsky. Sections 4.1 Spherically symmetric solutions.** Problems: 4.1.14-4.1.17.
5. **Mark Pinsky. Sections 4.3. Laplace equation, spherical coordinates.** Problems: 4.3.1, 4.3.2, 4.3.9, and 4.3.11.